

1.

Kirchhoff-Love $h = h(x, y)$ $\rho E \nu$

$w = w(x, y, t)$

$$\kappa_x = -\frac{\partial^2 w}{\partial x^2}, \quad \kappa_y = -\frac{\partial^2 w}{\partial y^2}, \quad \kappa_{xy} = -\frac{\partial^2 w}{\partial x \partial y} \quad (1)$$

2.

$h(x, y) D$

$$D(x, y) = \frac{Eh^3(x, y)}{12(1 - \nu^2)} \quad (2)$$

$M_x, M_y M_{xy}$

$$\begin{aligned} M_x &= -D(x, y) \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \\ M_y &= -D(x, y) \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \\ M_{xy} &= -D(x, y)(1 - \nu) \frac{\partial^2 w}{\partial x \partial y} \end{aligned} \quad (3)$$

3.

$q(x, y, t) - \rho h(x, y) \frac{\partial^2 w}{\partial t^2}$

$$\frac{\partial^2 M_x}{\partial x^2} + 2 \frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} + q(x, y, t) = \rho h(x, y) \frac{\partial^2 w}{\partial t^2} \quad (4)$$

$D(x, y)$

$$\begin{aligned} & \frac{\partial^2}{\partial x^2} \left[D \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \right] + 2 \frac{\partial^2}{\partial x \partial y} \left[D(1 - \nu) \frac{\partial^2 w}{\partial x \partial y} \right] \\ & + \frac{\partial^2}{\partial y^2} \left[D \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \right] + \rho h \frac{\partial^2 w}{\partial t^2} = q(x, y, t) \end{aligned} \quad (5)$$

$$\nabla^2 (D \nabla^2 w) - (1 - \nu) \left(\frac{\partial^2 D}{\partial x^2} \frac{\partial^2 w}{\partial y^2} - 2 \frac{\partial^2 D}{\partial x \partial y} \frac{\partial^2 w}{\partial x \partial y} + \frac{\partial^2 D}{\partial y^2} \frac{\partial^2 w}{\partial x^2} \right) + \rho h \frac{\partial^2 w}{\partial t^2} = q(x, y, t) \quad (6)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \quad D \nabla^4 w + \rho h \ddot{w} = q$$

$$4. (\mathcal{K} - \omega^2 \mathcal{M})W = Q$$

$$Q(x, y)e^{i\omega t} \quad W(x, y)e^{i\omega t} \quad (\mathcal{K} - \omega^2 \mathcal{M})W(x, y) = Q(x, y) \quad (7)$$

$$\mathcal{M} \mathcal{M}[W] = \rho h(x, y)W$$

$$\mathcal{K}[W] = \frac{\partial^2}{\partial x^2} \left[D \left(\frac{\partial^2 W}{\partial x^2} + \nu \frac{\partial^2 W}{\partial y^2} \right) \right] + 2 \frac{\partial^2}{\partial x \partial y} \left[D(1 - \nu) \frac{\partial^2 W}{\partial x \partial y} \right] + \frac{\partial^2}{\partial y^2} \left[D \left(\frac{\partial^2 W}{\partial y^2} + \nu \frac{\partial^2 W}{\partial x^2} \right) \right] \quad (8)$$

1.

$$\Pi(W)$$

$$\begin{aligned} \Pi(W) = & \frac{1}{2} \iint_{\Omega} D(x, y) \left[(\nabla^2 W)^2 - 2(1 - \nu) \left(\frac{\partial^2 W}{\partial x^2} \frac{\partial^2 W}{\partial y^2} - \left(\frac{\partial^2 W}{\partial x \partial y} \right)^2 \right) \right] dx dy \\ & - \frac{1}{2} \omega^2 \iint_{\Omega} \rho h(x, y) W^2 dx dy - \iint_{\Omega} F(x, y) W dx dy \end{aligned} \quad (9)$$

$$\delta \Pi = 0$$

2.

$$\mathbf{N} W(x, y) = \mathbf{N}(x, y) \mathbf{w}$$

$$\Pi(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T (\mathbf{K} - \omega^2 \mathbf{M}) \mathbf{w} - \mathbf{w}^T \mathbf{f} = \frac{1}{2} \mathbf{w}^T \mathbf{A} \mathbf{w} - \mathbf{w}^T \mathbf{f} \quad (10)$$

3. Hessian

$$\nabla_{\mathbf{w}} \Pi(\mathbf{w}) = \mathbf{A} \mathbf{w} - \mathbf{f} = \mathbf{0}$$

$$\text{Hessian } \mathbf{H} = \nabla_{\mathbf{w}}^2 \Pi(\mathbf{w}) = \mathbf{A}$$

(Adjoint Method)

1.

$$\mathbf{A}(\mathbf{p}) \mathbf{u} = \mathbf{f} \min_{\mathbf{p}} \mathcal{L}(\mathbf{u}, \mathbf{p})$$

2.

$$\frac{d\mathcal{L}}{d\mathbf{p}} = \frac{\partial \mathcal{L}}{\partial \mathbf{p}} + \frac{\partial \mathcal{L}}{\partial \mathbf{u}} \frac{d\mathbf{u}}{d\mathbf{p}} \quad (11)$$

$$\mathbf{p} \frac{d\mathbf{A}}{d\mathbf{p}} \mathbf{u} + \mathbf{A} \frac{d\mathbf{u}}{d\mathbf{p}} = \mathbf{0} \quad \frac{d\mathbf{u}}{d\mathbf{p}} = -\mathbf{A}^{-1} \left(\frac{d\mathbf{A}}{d\mathbf{p}} \mathbf{u} \right)$$

3.

$$\lambda$$

$$\mathbf{A}^T \boldsymbol{\lambda} = \left(\frac{\partial \mathcal{L}}{\partial \mathbf{u}} \right)^T \quad (12)$$

$$\frac{d\mathcal{L}}{d\mathbf{p}} = \frac{\partial \mathcal{L}}{\partial \mathbf{p}} - \boldsymbol{\lambda}^T \left(\frac{d\mathbf{A}}{d\mathbf{p}} \mathbf{u} \right) \quad (13)$$

4.

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